Banking Data Analytics

Mini Project Documentation

Source: Banking_Raw_Data_Mini_Project-2.xlsx | 08 November 2026

1. Executive Summary

This banking analytics mini-project examines 3,000 raw records across customer profiles, bank accounts, loan balances, transactions and service channels. The supplied workbook contains raw data, cleaning worksheets, pivot analysis and a dashboard sheet. The goal is to improve data quality and support banking performance reporting.

This report documents the workbook as provided; it does not assume that an empty or image-based dashboard is fully interactive.

2. Business Problem and Objectives

Banks need consistent customer, account and transaction data to understand operational performance and identify records requiring investigation.

• Standardize inconsistent names, categories and locations.

• Identify missing values, invalid ages and unusual negative balances or loan amounts.

• Summarize customer distribution, account types, transaction statuses and monthly transaction values.

• Build a foundation for interactive reporting in Excel or Power BI.

3. Source Workbook and Dataset

Workbook: Banking_Raw_Data_Mini_Project-2.xlsx. Raw_Data (2) has 3,000 data rows and 20 columns. Power_Query_Clean has 2,982 data rows; Clean_Data has 3,000 data rows.

The workbook includes five sheets: Power_Query_Clean, Raw_Data (2), Pivot Analysis, Dashboard and Clean_Data. The raw sheet is used for the descriptive metrics in this report.

Worksheet	Role
Raw_Data (2)	Original customer/account/transaction records
Clean_Data	Excel cleaning and validation columns
Power_Query_Clean	Standardized output for analysis
Pivot Analysis	Aggregated summaries
Dashboard	Dashboard layout and embedded objects

4. Data Dictionary

The raw data fields are grouped by business purpose.

Category	Fields
Customer	Customer_ID, Customer_Name, Age, Gender, City, Occupation
Account	Account_Number, Account_Type, Account_Open_Date, Branch, Balance_INR
Credit / loan	Credit_Score, Loan_Amount_INR

Category	Fields
Transaction	Transaction_ID, Transaction_Date, Transaction_Type, Transaction_Amount_INR, Transaction_Channel
Other	Transaction_Status, Is_Active

5. Data Preparation and Cleaning

The workbook provides both a Clean_Data worksheet with derived checks and a Power_Query_Clean worksheet. Cleaning should be documented as a repeatable sequence, not assumed to have corrected every anomaly.

• Import raw Excel data into Power Query and validate column types.

• Trim spaces, normalize capitalization and standardize variants such as Bengaluru/Bangalore and Salary/Salary Account.

• Check blank branch, gender, account and transaction fields.

• Validate age, account numbers, credit scores, negative balances and loan values.

• Review duplicates using Customer_ID and Transaction_ID as appropriate; do not automatically delete valid repeated transactions.

• Create review flags and refresh the cleaned output before building PivotTables.

6. Descriptive KPI Analysis

The following figures are calculated from the raw worksheet (3,000 rows). Monetary amounts are in Indian rupees and may include negative or unusual source values.

KPI	Raw-data result
Total records	3,000
Distinct customer IDs	2,998
Total account balance	INR 245,365,877.26
Total loan amount	INR 1,255,115,311.01
Total transaction amount (signed)	INR 37,885,915.69
Average credit score	575.8
Negative balances	745
Negative loan amounts	650

7. Account Type Distribution

Counts below reflect the raw labels for Account_Type; variant spellings may split equivalent categories.

Category	Records
Current	776
Fixed Deposit	760
Salary	693
Savings	683
Blank	53

Category	Records
FD	12
current	8
SAVINGS	7
Salary Account	5
saving	3

8. Transaction Channels

Counts below reflect the raw labels for Transaction_Channel; variant spellings may split equivalent categories.

Category	Records
Internet Banking	620
UPI	591
ATM	590
Mobile Banking	557
Branch	551
Blank	59
MOBILE banking	16
atm	7
InternetBanking	6
upi	3

9. Transaction Types

Counts below reflect the raw labels for Transaction_Type; variant spellings may split equivalent categories.

Category	Records
Transfer	632
Withdrawal	629
Payment	574
ATM Withdrawal	566
Deposit	564
WITHDRAWAL	10
ATM withdrawl	9
deposit	6
Bill Pay	5
fund transfer	5

10. Transaction Status

Status	Records
Successful	2271
Failed	311
Pending	236
Reversed	147
success	9
fail	7
reversed	7
in process	6
SUCCESSFUL	6

11. Pivot Table and Dashboard Design

The Pivot Analysis sheet includes summaries by branch, account type, transaction status and transaction month. These are suitable for reporting customer counts, transaction counts and monetary trends.

Suggested dashboard KPIs: total customers, total balance, loan portfolio, transaction volume, average credit score and failed transaction count. Suggested visuals: branch-wise customer count, account type mix, transaction status distribution and monthly transaction value.

• Connect slicers to PivotTables that share the same source or data model.

• Use Year, City, Branch and Account Type as practical slicers when supported by the dataset.

• For large monetary totals, display INR lakhs or crores with a clear unit label.

• Refresh data and all pivots before exporting or sharing.

12. Data Quality Findings

The raw dataset includes 745 negative balance entries and 650 negative loan amounts. These should be reviewed against business rules, rather than automatically removed.

Blank branch values: 66.

Some raw category labels and city spellings are inconsistent. The cleaned worksheets provide standardized and flagged fields, but validation is still necessary.

13. Business Insights and Recommendations

Account type and transaction channel distributions can inform branch operations and digital channel planning. Negative financial values and missing branches require investigation before using results for management decisions.

Signed transaction amounts must not be confused with total transaction volume or absolute cash movement. Customer counts, account counts and transaction counts should remain separate business measures.

• Prioritize flagged records and document correction rules.

• Use a common date calendar for year and month comparisons.

• Create a data dictionary and validate KPI definitions with stakeholders.

• Schedule regular refresh and reconciliation checks.

14. Tools and Skills Demonstrated

Microsoft Excel, Power Query, data validation, data cleaning, PivotTables, KPI reporting, banking domain analysis and dashboard design.

Potential Power BI extension: load the cleaned table, create DAX measures, model date relationships and publish an interactive report.

15. Project Conclusion

The project demonstrates an end-to-end banking analytics workflow from raw data to cleaned data, aggregated PivotTables and dashboard planning. Reliable insights depend on consistent categories, explicit KPI definitions and review of anomalous source records.

16. Interview-Ready Project Explanation

I worked on a Banking Data Analytics mini-project using Excel and Power Query. I cleaned customer, account and transaction records, standardized categories, reviewed missing and negative values, and summarized the dataset with PivotTables. I focused on banking KPIs such as customer count, account balance, loan amount, transaction trends and transaction status. The project improved my practical skills in data cleaning, reporting and business analysis.